

Image Processing & Recognition: Lecture 9 Review

Welcome to Level 9! We are now adding the fourth dimension: **Time**. This lecture covers how to recognize actions (e.g., "swimming", "jumping") in video data.

Part 1: What are you learning? (The "Human" Summary)

1. The Challenge: The Third Dimension

- **Images:** Height $\times$ Width $\times$ Channels ($H \times W \times C$).
- **Videos:** Frames $\times$ Height $\times$ Width $\times$ Channels ($T \times H \times W \times C$).
- **Problem:** Videos are massive. A simple dense network would require impossible amounts of memory.
- **Solution:** We process short clips (snippets) or use specialized architectures that handle time efficiently.

2. The Architectures

- **Two-Stream Networks:** Separates the problem into two distinct neural networks:
 1. **Spatial Stream:** Looks at a single frame (RGB) to determine *what* objects are there.
 2. **Temporal Stream:** Looks at "Optical Flow" (motion vectors) to determine *how* things are moving.
 - The results are averaged at the end.
- **C3D (3D Convolutional Networks):**
 - Instead of flat 2D squares filters, we use **3D Cubic filters** ($d \times k \times k$).
 - These filters slide across height, width, *and* time, capturing motion and appearance simultaneously.
- **TSN (Temporal Segment Networks):**
 - Solves the "Long Video" problem. It chops a long video into segments (e.g., 3 parts), picks one random snippet from each, processes them, and averages the result. This captures long-range context without processing every single frame.
- **I3D (Inflated 3D):**
 - Takes a famous 2D image classifier (like Inception) and "inflates" the filters to 3D. It allows us to use pre-trained image knowledge for video tasks.
- **SlowFast:**
 - Inspired by biology (retinal cells).
 - **Slow Pathway:** Looks at low frame rate (detail) to understand the scene.

- **Fast Pathway:** Looks at high frame rate (motion) but with fewer channels to understand movement quickly.

Part 2: The Generated Solved Problem (Step-by-Step)

Since the PDF did not have a specific exercise, I have designed one based on the core mathematical operation introduced in this lecture: **3D Convolution** (calculating output dimensions).

Exercise 9

We want to calculate the output dimensions of a **3D Convolutional Layer**. This is critical for understanding architectures like C3D.

Given:

- **Input Volume (V_{in}):** A video clip consisting of 16 frames, with a resolution of 112×112 . Dimensions: $T_{in} = 16, H_{in} = 112, W_{in} = 112$. (We assume 1 input channel for simplicity, or that we are calculating the spatial/temporal map size regardless of channels).
- **Kernel (K):** A 3D filter of size $3 \times 3 \times 3$. Dimensions: $d = 3$ (temporal depth), $k = 3$ (spatial height/width).
- **Stride (s):** 1 (spatially and temporally).
- **Padding (p):** 0 (valid convolution).

Task: Calculate the dimensions of the Output Volume ($T_{out}, H_{out}, W_{out}$).

Solution:

Step 1: Identify the Formula The formula for the output dimension of a convolution is the same for time as it is for space.

$$Dim_{out} = \left\lfloor \frac{Dim_{in} - K + 2P}{S} \right\rfloor + 1$$

- **Source: Page 41** provides the general formula for dilated convolution output, which simplifies to the standard formula when dilation rate $r = 1$: $\lfloor (n - k + p + s)/s \rfloor$. (Note: standard formulations usually write this as $(n - k + 2p)/s + 1$). Let's stick to the logic provided in the earlier convolution lecture (Lecture 3, Page 30) extended to 3D: Output = Input - Kernel + 1 (for stride 1, pad 0).

Step 2: Calculate Temporal Dimension (T_{out})

- $T_{in} = 16$
- Kernel Depth $d = 3$
- $T_{out} = 16 - 3 + 1 = 14$.

Step 3: Calculate Spatial Dimensions (H_{out}, W_{out})

- $H_{in} = 112$

- Kernel Size $k = 3$
- $H_{out} = 112 - 3 + 1 = 110.$
- $W_{out} = 112 - 3 + 1 = 110.$

Step 4: Final Output Volume The output volume is $14 \times 110 \times 110$.

- **Source: Page 33** describes the "Common network settings" for C3D, mentioning how input clips ($3 \times 16 \times 128 \times 171$) are processed. While our specific numbers differ, the *mechanism* of 3D volume reduction follows the logic described in **Page 27** ("3D convolution is achieved by convolving a 3D kernel to the cube").

Part 3: Your Turn (Interactive Exercises)

Practice Problem 1: Parameter Counting (3D Conv)

Let's calculate the "cost" of a 3D filter versus a 2D filter.

Setup:

- **2D Kernel:** 3×3 . Input Channels: 64. Output Channels: 128.
- **3D Kernel:** $3 \times 3 \times 3$. Input Channels: 64. Output Channels: 128.

Task: Calculate the number of weights (parameters) for both. Ignore bias.

Solving Explanation:

1. **2D Parameters:** $3 \times 3 \times 64 \times 128 = 9 \times 8192 = 73,728$.
2. **3D Parameters:** $3 \times 3 \times 3 \times 64 \times 128 = 27 \times 8192 = 221,184$.
 - (Note: 3D convolution is 3x more expensive in parameters here because the temporal depth is 3).
- **Source: Page 36** discusses the number of parameters in C3D networks and compares depth-1 vs depth-7 networks.

Practice Problem 2: SlowFast Frame Sampling

Setup: You are designing a SlowFast network (described on **Page 72**).

- **Raw Clip Length:** 64 frames.
- **Slow Pathway Stride (τ):** 16 (samples 1 frame every 16).
- **Alpha (α):** 8 (Fast pathway has 8x higher frame rate).

Task: Calculate how many frames enter the **Slow Pathway** and how many enter the **Fast Pathway**.

Solving Explanation:

1. **Slow Pathway Frames (T_{slow}):** $64/16 = 4$ frames.
2. **Fast Pathway Stride:** The stride is $\tau/\alpha = 16/8 = 2$. (Samples 1 frame every 2).

3. **Fast Pathway Frames (T_{fast}):** $64/2 = 32$ frames.

- **Source:** Page 72 states: "Fast pathway works with a small temporal stride of τ/α ."